

Teaching Generative AI: Deploying ComfyUI to the Classroom with Docker and Cloud-based GPUs

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Abstract

This one-hour workshop presents a reproducible approach to teaching generative AI through cloud-first deployment and visual programming. Drawing from the School of Visual Arts’ “Innovating with Generative AI” course, we demonstrate how ComfyUI, a node-based interface for diffusion models, can be deployed at classroom scale using Docker containers and platforms such as Runpod. Participants will learn strategies for teaching text-to-image, image-to-3D, and multimodal workflows without requiring local GPU hardware. Validated over three semesters, this framework lowers technical barriers and enables educators to integrate professional-grade AI tools into diverse curricula.

CCS Concepts

• **Software and its engineering** → **Visual languages**; • **Computing methodologies** → *Mesh models*; *Neural networks*; *Image processing*.

Keywords

AI Education, ComfyUI, Generative AI, Docker, Cloud Computing, Visual Programming, Pedagogical Innovation, Educators Forum, Diffusion Models

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1 Author Biographies

Michael Gold is an entrepreneur and researcher who builds and invests in the 3D, AI, and open science ecosystem. His work blends 3D, generative AI, and spatial interaction across academia, startups, and venture, with showcases at SIGGRAPH, Sundance, Art Basel, and SXSW. He is a faculty member at the School of Visual Arts

and the founder of Horseman.AI, where he is building a modular system for embodied agents.

2 Introduction

2.1 The Critical Gap in AI Education

Current computer graphics education faces an unprecedented challenge: while industry rapidly adopts generative AI tools, academic curricula lag behind due to infrastructure barriers. Traditional approaches fail because:

- **Hardware Inequality:** Students without high-end GPUs cannot participate meaningfully
- **Installation Complexity:** Setting up modern AI tools requires expert-level system administration
- **Rapid Tool Evolution:** Academic procurement cycles cannot keep pace with monthly model releases

Our solution addresses each barrier through cloud-first infrastructure, containerized deployment, and portfolio-based assessment.

Generative AI represents a paradigm shift in creative education, where artificial intelligence transforms from a theoretical concept into a practical, hands-on creative medium. At the School of Visual Arts, our “Innovating with Generative AI” course has pioneered an approach that democratizes access to professional-grade AI tools through cloud-first deployment strategies and visual programming interfaces.

This workshop demonstrates how educators can implement a complete generative AI curriculum using ComfyUI [Contributors 2025], a powerful node-based visual interface that eliminates coding barriers while maintaining full creative control. Our methodology addresses the fundamental challenge facing AI education: how to provide students with immediate access to cutting-edge generative models without requiring expensive local hardware or complex technical setup.

Through our containerized approach using the comfy3d project and cloud platforms like Runpod, we’ve successfully deployed interactive AI workshops for classes of 20+ students simultaneously, each accessing GPU-accelerated diffusion models through their web browsers. This session shares our proven educational framework, technical infrastructure, and pedagogical insights from teaching the next generation of AI-native creators. [Gold and Chang 2025]

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Table 1: Educational Technology Stack: Tools and Infrastructure Used in Teaching Generative AI

Tool/Technology	Type	Educational Use	Learning Objective	Pedagogical Value
ComfyUI	Interface	Visual node-based programming	Creative workflow design	Reduces technical barriers, intuitive learning
Comfy3D Container	Infrastructure	Pre-compiled containerization	Zero-setup deployment	Eliminates installation complexity
Runpod Cloud Template	Infrastructure	GPU-accelerated cloud deployment	Hardware democratization	Access regardless of student hardware
HF Model Downloader	Tool	Automated model management	Resource organization & licensing	Streamlined model access, legal compliance
Docker Containerization	Infrastructure	Consistent environment deployment	Technical reproducibility	Standardized learning environment

3 Workshop Structure

This one-hour educators forum session demonstrates our complete teaching methodology through four integrated components:

3.1 Educational Philosophy & Cloud-First Strategy (0:00 – 0:15)

We begin by presenting our core teaching philosophy: “Learn by making, not just watching.” This approach prioritizes creative experimentation with immediate feedback over traditional lecture-based instruction. We demonstrate our cloud-first deployment strategy, showing how the comfy3d Docker container enables zero-setup access to GPU-accelerated AI workflows through platforms like Runpod.

Key Pedagogical Insights:

- Visual, node-based programming reduces technical barriers
- Cloud deployment ensures equitable access regardless of student hardware
- Immediate visual feedback accelerates learning and maintains engagement
- Modular workflows teach both creative and technical thinking

3.2 Core Educational Workflows (0:15 – 0:35)

Live demonstration of foundational teaching modules from our SVA curriculum:

- **Text-to-Image Foundations:** Teaching diffusion concepts through interactive KSampler experimentation
- **Visual Control Systems:** Workflows for understanding conditional generation
- **Creative Problem-Solving:** Exercises that develop artistic and technical intuition

Each workflow is designed to be completed within a single class session while building toward more complex multi-week projects.

3.3 Advanced Multi-Modal Teaching (0:35 – 0:50)

Demonstration of cutting-edge pedagogical applications that represent the frontier of AI education:

- **Image-to-3D Pipelines:** Using Hunyuan3D-2[Zhao and et al 2025] and Trellis[Xiang et al. 2024] to teach spatial reasoning and 3D content creation
- **Character Consistency:** LoRA training workflows that teach students about model fine-tuning and creative control
- **Video Generation:** Temporal diffusion concepts through Wan2.2[Wan and et al 2025] and commercial API integration

3.4 Implementation Strategies & Q&A (0:50 – 1:00)

Practical guidance for educators implementing similar programs:

- Infrastructure requirements and cost considerations
- Curriculum integration strategies for different program types
- Assessment methods for creative AI projects
- Student outcomes and career preparation insights

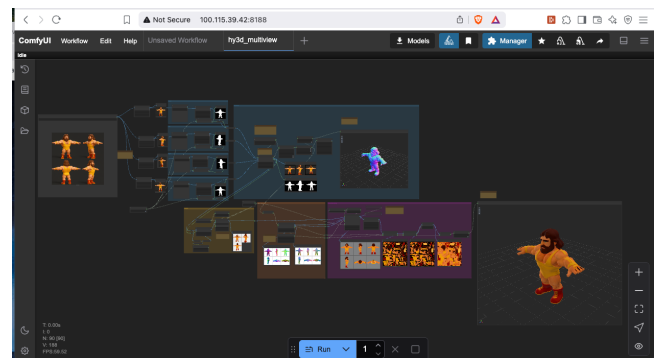


Figure 1: Students using ComfyUI in a cloud-deployed classroom environment, running Hunyuan3D-2 workflows for image-to-3D generation. The visual node-based interface enables intuitive learning while providing access to cutting-edge AI models without requiring local GPU hardware.

4 Educational Technology Stack

Our pedagogical approach is built on three core technological pillars that ensure accessibility, scalability, and educational effectiveness.

4.1 Key Technical Innovations

Our educational deployment solves three critical technical challenges:

- (1) **The "Dependency Hell" Problem:** Traditional AI education requires students to navigate complex CUDA compilations, often consuming entire class sessions. Our pre-compiled wheels (Table 2) remove these barriers.
- (2) **The "Model Management" Challenge:** With 50+ models required for a complete curriculum, students typically spend hours downloading and organizing files. Our automated downloader reduces this to a single click.
- (3) **The "Environment Drift" Issue:** Student workstations with different OS versions, Python installations, and GPU drivers create inconsistent learning experiences. Containerization ensures every student works in an identical environment.

4.2 Comfy3D: Production-Ready Containerization

The comfy3d project [Gold 2025a] provides a pre-compiled Docker container that eliminates the traditional barriers to deploying complex 3D diffusion models in educational settings.

Quick Deployment:

```
git clone https://github.com/michaelgold/comfy3d
cd comfy3d
docker-compose up
```

This container includes:

- **Precompiled CUDA Extensions:** Differential rasterization and mesh processing libraries are pre-built, eliminating compilation errors and setup complexity
- **Integrated 3D Pipeline Support:** Native compatibility with Hunyuan3D-2, Trellis, and other cutting-edge 3D generation models
- **Educational Model Presets:** Curated selection of models optimized for classroom use and learning progression
- **Sample Workflow Collection:** Pre-loaded JSON workflow files for text-to-image, image-to-3D, video generation, and multi-modal pipelines, allowing students to experiment with known good starting points immediately
- **Automated Dependency Management:** All Python packages, custom nodes, and system dependencies are pre-configured
- **Flexible Persistent Storage:** Intelligent workspace initialization that supports both local Docker volumes and cloud network storage

Educational Impact: Because these wheels were pre-compiled, we were able to onboard a classroom of students in under 10 minutes without a single dependency failure. This eliminated the traditional “first week lost to installation problems” that plagues technical coursework, allowing students to focus immediately on creative experimentation and learning outcomes.

Persistent Workspace Management: The container automatically detects and configures persistent storage for educational content. When deployed with the `init-workspace` flag, the system:

- **Auto-detects Storage Type:** Uses `/runpod-volume` for cloud deployments or `/workspace` for local containers
- **Preserves Student Work:** Symlinks critical directories (models, custom_nodes, output, input, workflows) to persistent storage
- **Enables Seamless Migration:** Students can move between different container instances while retaining their work and installed models
- **Supports Shared Resources:** Multiple students can share common models while maintaining individual workspaces

This storage architecture ensures that educational content persists across container restarts and enables flexible deployment across different cloud platforms without data loss.

This approach transforms what traditionally required hours of technical setup into a single Docker command, enabling educators to focus on teaching rather than troubleshooting installations.

4.3 Cloud-First Deployment with Runpod

Runpod templates provide a streamlined way to deploy educational environments. Using our comfy3d template (<https://console.runpod.io/deploy?template=j75puttb7q>), a single educator can provision 20+ GPU instances simultaneously, each running the complete ComfyUI environment accessible through secure web interfaces.

Key Educational Benefits:

- **Hardware Democratization:** Students with basic laptops gain access to A40/A100 GPU acceleration
- **Consistent Environment:** Every student works with identical software versions and model access
- **Cost Efficiency:** Pay-per-hour GPU usage (approximately \$0.22/hour per student) eliminates large capital investments
- **Instant Scalability:** Workshop deployment scales from 5 to 50 students without infrastructure changes

4.4 Automated Model Management

The ComfyUI-HF-Model-Downloader [Gold 2025b] provides a web-based interface for managing the complex landscape of AI model licensing and distribution. This tool:

- **Simplifies Authentication:** Automated Hugging Face token management for license-gated models
- **Batch Downloads:** One-click installation of complete model suites (FLUX, Hunyuan3D-2, Wan2.2, QWEN Image, etc.)
- **Version Control:** Ensures consistent model versions across all student instances
- **Legal Compliance:** Automatic license acceptance and tracking for educational use

5 Pedagogical Innovation: Teaching Generative AI

Our approach to AI education is grounded in the philosophy that teaching generative AI is fundamentally similar to teaching procedural graphics. Students must develop an intuition for working with noise. And computer graphics education, built around deterministic algorithms and precise mathematical operations, must evolve to embrace the probabilistic, iterative nature of generative AI.

Table 2: Pre-Compiled CUDA Dependencies: Eliminating Installation Barriers for Educational Success

Package	Known Installation Challenge
pytorch3d	Complicated build process, often fails on student workstations with inconsistent CUDA versions
flash-attn	Requires CUDA 12+, frequent compiler mismatches across different systems
nvdiffrast	Complex CUDA compilation, requires specific driver versions and build tools
spconv	Sparse convolution builds fail frequently, especially on non-Linux systems
diff-gaussian-rasterization	Cutting-edge research code with unstable build requirements
pytorch_scatter	Scatter operation compilation errors common in mixed environments
hy3dgen-rasterizer	Custom CUDA kernels require specific PyTorch versions and build configurations

5.1 Visual Programming as Creative Thinking

ComfyUI’s node-based interface serves as more than a technical tool, it becomes a visual language for thinking about creative processes. Students learn to decompose complex creative goals into modular, reusable components, developing both artistic and computational thinking simultaneously.

Key Learning Outcomes:

- **Systems Thinking:** Understanding how creative processes can be broken into discrete, interconnected steps
- **Iterative Design:** Embracing uncertainty and refinement as core creative methodologies
- **Modular Creativity:** Building reusable creative components that can be recombined in novel ways
- **Technical Literacy:** Gaining fluency with AI concepts without requiring traditional programming skills

5.2 Hands-On Learning with Immediate Feedback

Our curriculum follows a progressive complexity model (Table 3), scaffolding from basic text-to-image foundations to advanced multi-modal pipelines and independent projects.

Table 3: Progressive Complexity Model for Generative AI Curriculum

Weeks	Focus	Example Techniques
1–2	Foundations	Text-to-image workflows, prompt engineering
3–6	Multi-modal	Inpainting, image-to-image, ControlNet
7–10	Advanced	LoRA training, 3D generation, video synthesis
11–15	Projects	Independent pipelines, API integration

Each phase builds technical complexity while maintaining creative focus, ensuring students remain engaged and productive throughout the learning process.

5.3 Assessment Through Creative Portfolio Development

Traditional computer science assessment methods (exams, coding assignments) are inadequate for evaluating creative AI competency. Our approach emphasizes portfolio development and creative project completion:

- **Process Documentation:** Students maintain visual journals of their workflow development
- **Technical Mastery:** Demonstrated through increasingly complex multi-modal projects
- **Creative Innovation:** Evaluated through original applications of learned techniques
- **Collaboration:** Peer learning through workflow sharing and collaborative debugging

6 Educational Outcomes and Student Success

Over three semesters of teaching “Innovating with Generative AI” at the School of Visual Arts, we have documented significant learning outcomes that demonstrate the effectiveness of our cloud-first, visual programming approach.

6.1 Qualitative Learning Outcomes

Students demonstrate mastery across multiple dimensions:

Creative Competencies:

- Advanced prompt engineering and creative direction skills
- Intuitive understanding of diffusion model behavior and conditioning
- Ability to combine multiple AI tools into coherent creative pipelines
- Development of personal artistic voice within AI-generated content

Technical Skills:

- Proficiency with node-based visual programming concepts
- Understanding of cloud computing and containerization basics
- Experience with API integration and workflow automation
- Knowledge of model training concepts through LoRA implementation

6.2 Career Preparation and Industry Relevance

Our graduates enter the workforce with immediately applicable skills:

- **Creative Industry Applications:** Advertising, film VFX, game development, architectural visualization
- **Technical Roles:** AI product development, creative tool design, workflow optimization

6.3 Peer Teaching and Independent Exploration

A key outcome of this course has been students' ability to extend learning beyond the formal curriculum. All students independently implemented workflows and models not covered in class by following external tutorials, and then taught these new pipelines to their peers through live demonstrations and shared documentation. This reflects a successful transfer of learning: students not only mastered the taught material, but also developed the confidence and collaborative habits to explore new models, adapt workflows, and sustain peer-to-peer instruction. Such outcomes indicate a self-sustaining learning culture, where students evolve from learners into contributors within the generative AI community.

7 Workshop Learning Objectives

By the end of this workshop, participants will be able to:

- Deploy a complete ComfyUI environment for 20+ students in under 15 minutes
- Design progressive curriculum modules that build from basic text-to-image to advanced 3D workflows
- Implement cost-effective cloud infrastructure for AI education (estimated \$0.22 per student per hour)
- Assess student progress in generative AI through portfolio-based evaluation methods
- Troubleshoot common deployment and pedagogical challenges in AI education

Take-Away Resources:

- Complete semester curriculum template with 15 progressive workshop modules
- Pre-configured Docker containers and cloud deployment scripts
- Student assessment rubrics and portfolio evaluation frameworks
- Cost analysis spreadsheet for institutional budget planning

8 Implementation Guide for Educators

This workshop provides attendees with a complete implementation roadmap for deploying similar programs at their own institutions.

8.1 Technical Requirements and Costs

Minimum Infrastructure:

- Cloud computing account (Runpod, AWS, or private cloud with A40 GPU or equivalent)
- Budget: \$2 per student per 3-hour session
- Instructor laptop with reliable internet connection
- Classroom with individual workstations or personal laptops

Optional Enhancements:

- Institutional cloud computing credits or partnerships
- Local GPU server for reduced ongoing costs
- Custom domain and branding for student instances

8.2 Curriculum Integration Strategies

Our methodology adapts to diverse educational contexts:

Computer Graphics Programs:

- Replace traditional modeling exercises with AI-assisted rapid prototyping
- Integrate as advanced elective focusing on emerging technologies
- Use as bridge between 2D and 3D coursework

Art and Design Programs:

- Focus on creative applications and artistic development
- Emphasize workflows over one-shot prompts
- Integrate with traditional media and mixed-media approaches

Computer Science Programs:

- Emphasize API integration and workflow automation
- Include custom node development and Python programming
- Connect to machine learning and neural network coursework

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Table 4: CUDA Architecture Support and GPU Recommendations for Educational Deployment

CUDA Arch	Consumer GPUs	Professional GPUs	VRAM Range	Suitability
6.1 (Pascal)	GTX 1080, 1080 Ti	P40, P100	8–12 GB	Basic
7.0 (Volta)	—	V100	16 GB	Limited 3D
7.5 (Turing)	RTX 2060–2080	T4	6–16 GB	Entry-level
8.0 (Ampere)	RTX 3080, 3090	A100	10–40 GB	Standard
8.6 (Ampere)	RTX 3060, 3070	A30, A40	8–48 GB	Recommended
8.9 (Ada)	RTX 4080, 4090	L40, L40S	16–48 GB	Optimal
9.0 (Hopper)	RTX 5080, 5090	H100, H200	24–80 GB	Future-ready
12.0 (Blackwell) ¹	—	GB200, GB100	48–192 GB	Next-gen
Recommendation: 24+ GB VRAM for full 3D workflows (e.g., A40, RTX 4090, RTX 5090)				